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Audio Capture with Multiple Devices

Abstract

In one implementation, a method of visualizing a combined audio pick-up pattern is performed at a first device in a physical environment, the first device including a display, one or more processors, and non-transitory memory. The method includes determining a first audio pick-up pattern of the first device. The method includes determining one or more second audio pick-up patterns of a respective one or more second devices. The method includes determining a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns. The method includes displaying, on the display, a representation of the combined audio pick-up pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/212,488, filed on Jun. 21, 2023, which claims priority to U.S. Provisional Patent No. 63/356,624, filed on Jun. 29, 2022, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to determining an audio emission pattern of a physical sound source using multiple devices.

BACKGROUND

[0003] A virtual object in an extended reality (XR) environment may be an audio emitter object that emits sound in an XR environment with a volume that depends on the relative location and/or orientation of the virtual object and the user in the XR environment according to an audio emission pattern.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] So that the present disclosure can be understood by those of ordinary skill in the art, a more detailed description may be had by reference to aspects of some illustrative implementations, some of which are shown in the accompanying drawings.

[0005] FIGS. 1A-1C illustrate a first physical environment.

[0006] FIGS. 2A-2B illustrate a second XR environment.

[0007] FIG. 3 is a flowchart representation of a method of visualizing a combined audio pick-up pattern in accordance with some implementations.

[0008] FIG. 4 is a flowchart representation of a method of determining an audio emission pattern in accordance with some implementations.

[0009] FIG. 5 is a block diagram of an electronic device in accordance with some implementations.

[0010] In accordance with common practice the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

SUMMARY

[0011] Various implementations disclosed herein include devices, systems, and methods for visualizing a combined audio pick-up pattern. In various implementations, the method is performed at a first device in a physical environment, the first device having a display, one or more processors, and non-transitory memory. The method includes determining a first audio pick-up pattern of the first device. The method includes determining one or more second audio pick-up patterns of a respective one or more second devices. The method includes determining a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns. The method includes displaying, on the display, a representation of the combined audio pick-up pattern.

[0012] Various implementations disclosed herein include devices, systems, and methods for determining an audio emission pattern. In various implementations, the method is performed at a

first device at a first location, the first device having a microphone, one or more processors, and non-transitory memory. The method includes obtaining, via the microphone, first audio of a sound source. The method includes receiving, from one or more second devices, one or more second audio of the sound source. The method includes determining one or more second locations of the one or more second devices. The method includes determining an audio emission pattern of the sound source based on the first audio data, the one or more second audio data, and the one or more second locations, wherein the audio emission pattern of the sound source indicates a sound level at various locations relative to the sound source.

[0013] In accordance with some implementations, a device includes one or more processors, a non-transitory memory, and one or more programs; the one or more programs are stored in the non-transitory memory and configured to be executed by the one or more processors. The one or more programs include instructions for performing or causing performance of any of the methods described herein. In accordance with some implementations, a non-transitory computer readable storage medium has stored therein instructions, which, when executed by one or more processors of a device, cause the device to perform or cause performance of any of the methods described herein. In accordance with some implementations, a device includes: one or more processors, a non-transitory memory, and means for performing or causing performance of any of the methods described herein.

DESCRIPTION

[0014] A physical environment refers to a physical place that people can sense and/or interact with without aid of electronic devices. The physical environment may include physical features such as a physical surface or a physical object. For example, the physical environment corresponds to a physical park that includes physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment such as through sight, touch, hearing, taste, and smell. In contrast, an extended reality (XR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic device. For example, the XR environment may include augmented reality (AR) content, mixed reality (MR) content, virtual reality (VR) content, and/or the like. With an XR system, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the XR environment are adjusted in a manner that comports with at least one law of physics. As an example, the XR system may detect movement of the electronic device presenting the XR environment (e.g., a mobile phone, a tablet, a laptop, a head-mounted device, and/or the like) and, in response, adjust graphical content and an acoustic field presented by the electronic device to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), the XR system may adjust characteristic(s) of graphical content in the XR environment in response to representations of physical motions (e.g., vocal commands).

[0015] There are many different types of electronic systems that enable a person to sense and/or interact with various XR environments. Examples include head-mountable systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head-mountable system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head-mountable system may be configured to accept an external opaque display (e.g., a smartphone). The head-mountable system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head-mountable system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of

images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, uLEDs, liquid crystal on silicon, laser scanning light sources, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In some implementations, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

[0016] Numerous details are described in order to provide a thorough understanding of the example implementations shown in the drawings. However, the drawings merely show some example aspects of the present disclosure and are therefore not to be considered limiting. Those of ordinary skill in the art will appreciate that other effective aspects and/or variants do not include all of the specific details described herein. Moreover, well-known systems, methods, components, devices, and circuits have not been described in exhaustive detail so as not to obscure more pertinent aspects of the example implementations described herein.

[0017] As noted above, a virtual object in an extended reality (XR) environment may be an audio emitter object that emits sound in an XR environment with a volume that depends on the relative location and/or orientation of the virtual object and the user in the XR environment according to an audio emission pattern.

[0018] In various implementations, it may be beneficial for a virtual object to have an audio emission pattern substantially similar to a physical version of the virtual object. Determining the audio emission pattern of a physical object can be difficult and/or require highly controlled conditions. Accordingly, in various implementations described herein, the audio emission pattern of a physical object is determined using multiple electronic devices, such as smartphones, tablets, and/or head-mounted devices.

[0019] FIG. 1A illustrates a first physical environment **100** at a first time. The first physical environment **100** is associated with a three-dimensional environment coordinate system in which each point in the environment coordinate system is associated with an x-coordinate, a y-coordinate, and a z-coordinate. The first physical environment **100** includes a primary electronic device **110** including a primary display **111**. On an opposite side of the primary display **111**, the primary electronic device **110** includes a primary camera (not shown) having a primary camera pose (e.g., location and orientation) in the environment coordinate system. On the opposite side of the primary display **111**, the primary electronic device **110** includes a primary microphone (not shown) at a primary microphone pose in the environment coordinate system. The primary microphone has a primary audio pick-up pattern.

[0020] The audio pick-up pattern for a microphone indicates the directionality of the microphone. For example, in various implementations, a polar audio pick-up pattern for a microphone indicates, at each angle, the ratio of the volume of sound recorded by the microphone to the volume of a sound source at that angle. As another example, in various implementations, a local Cartesian audio pick-up pattern for a microphone indicates, at each point in a microphone coordinate system having an origin at the location of the microphone, the ratio of the volume of sound recorded by the microphone to the volume of a sound source at that point. In various implementations, the local Cartesian audio pick-up pattern can be determined from the polar audio pick-up pattern using an inverse-square law. As another example, a global Cartesian audio pick-up pattern for a microphone indicates, at each point in the environment coordinate system, the ratio of the volume of sound recorded by the microphone to the volume of a sound source at that point. In various implementations, the global Cartesian audio pick-up pattern can be determined from the local Cartesian audio pick-up pattern using a transform based on the pose of the microphone in the environment coordinate system.

[0021] The first physical environment **100** includes a first secondary electronic device **120A**. The

first secondary electronic device **120A** includes a first secondary microphone **121A** at a first secondary microphone pose in the environment coordinate system. The first secondary microphone **121A** has a first secondary audio pick-up pattern. The first secondary electronic device **120A** includes a first secondary display **122A** displaying a QR code **123**. The first physical environment **100** includes a second secondary electronic device **120B**. The second secondary electronic device **120A** includes a second secondary microphone **121A** at a second secondary microphone pose in the environment coordinate system. The second secondary electronic device **120A** has a second secondary audio pick-up pattern.

[0022] The first physical environment **100** includes a cricket **130**, a physical sound source.

[0023] The primary display **111** displays a first XR environment **140**. The first XR environment **140** includes a physical environment representation **150** of a portion of the first physical environment **100** augmented with a virtual pick-up representation **161**. In various implementations, the physical environment representation **150** is generated based on an image of the first physical environment **100** captured with the primary camera of the primary electronic device **110** having a field-of-view directed toward the first physical environment **100**. Accordingly, the physical environment representation **150** includes a first secondary electronic device representation **151A** of the first secondary electronic device **120A**, a second secondary electronic device representation **151B** of the second secondary electronic device **120B**, and a cricket representation **152** of the cricket **130**.

[0024] The XR environment **140** includes a virtual pick-up representation **161**. To display the virtual pick-up representation **161**, the primary electronic device **161** determines one or more sets of three-dimensional coordinates in the environment coordinate system of the virtual pick-up representation **161**. The primary electronic device **110** determines one or more locations on the primary display **111** (e.g., one or more sets of two-dimensional coordinates in a display coordinate system) corresponding to the one or more sets of three-dimensional coordinates in the environment coordinate system using a transform based on the primary camera pose (e.g., extrinsic parameters of the primary camera and, in various implementations, intrinsic parameters of the primary camera, such as the focal length, field-of-view, resolution, etc.). Then, the primary electronic device **161** displays the virtual pick-up representation at the locations on the primary display **111**.

[0025] The primary electronic device **110** determines the one or more sets of three-dimensional coordinates in the environment coordinate space for the virtual pick-up representation **161** based on the poses of the microphones (e.g., the primary microphone pose, the first secondary microphone pose, and the second secondary microphone pose) and their audio pick-up patterns (e.g., the primary audio pick-up pattern, the first secondary audio pick-up pattern, and the second secondary audio pick-up pattern).

[0026] The sound of a sound source recorded by multiple microphones can be combined using appropriate synchronization and processing to generate a combined sound. A combined audio pick-up pattern for multiple microphones can be defined as, at each point in the environment coordinate system, the ratio of the volume of the combined sound to the volume of the sound source. In various implementations, the combined audio pick-up pattern can be determined by combining (e.g., adding) the global Cartesian audio pick-up patterns of each of the multiple microphones.

[0027] Accordingly, in various implementations, the primary electronic device **110** determines the combined audio pick-up pattern for the primary microphone, the first secondary microphone **121A**, and the second secondary microphone **121B** based on the poses of the microphones and their audio pick-up patterns. In various implementations, the primary electronic device **110** determines the primary microphone pose using an inertial measurement unit (IMU) of the primary electronic device **110**. In various implementations, the primary electronic device **110** determines the secondary microphone poses by detecting the secondary electronic devices **120A-120B** in the image of the physical environment **100**. In various implementations, the primary electronic device **110** determines the secondary microphone poses by receiving pose data from the secondary

electronic devices **120A-120B** (which may be based on IMUs of the secondary electronic devices **120A-120B**). In various implementations, the primary electronic device **110** determines the secondary microphone poses by reading machine-readable code displayed by the secondary electronic devices **120A-120B**, such as the QR code **123** displayed by the first secondary electronic device **120A**.

[0028] In various implementations, the primary electronic device **110** determines the primary audio pick-up pattern by reading data from a memory of the primary electronic device **110**. In various implementations, the primary electronic device **110** determines the secondary audio pick-up patterns by detecting the secondary electronic devices **120A-120B** in the image of the physical environment **100**, classifying each of the secondary electronic devices **120A-120B** as a device type, and obtaining data from a memory of the primary electronic device **110** or a remote database associating device types with audio pick-up patterns. In various implementations, the primary electronic device **110** determines the secondary audio pick-up patterns by receiving data from the secondary electronic devices **120A-120B**. In various implementations, the primary electronic device **110** determines the secondary audio pick-up patterns by reading machine-readable code displayed by the secondary electronic devices **120A-120B**, such as the QR code **123** displayed by the first secondary electronic device **120A**.

[0029] In various implementations, the primary electronic device **110** determines one or more sets of coordinates in the environment coordinate system for the virtual pick-up representation **161** at which the combined audio pick-up pattern is a threshold value. In various implementations, the primary electronic device **110** determines one or more sets of coordinates in the environment coordinate system for the virtual pick-up representation **161** at which the combined audio pick-up pattern is greater than or equal to a threshold value.

[0030] In various implementations, the threshold value is a default value. In various implementations, the threshold value is based on the number of microphones. In various implementations, the threshold value is based on the device types of the primary electronic device **110** and the secondary electronic devices **120A-120B**. In various implementations, the threshold value is based on user input, e.g., setting or changing the threshold value.

[0031] In various implementations, the audio pick-up patterns (and, resultantly, the combined audio pick-up pattern) are frequency-dependent. Accordingly, in various implementations, the primary electronic device **110** determines one or more sets of coordinates in the environment coordinate system for the virtual pick-up representation **161** at which the combined audio pick-up pattern at a particular frequency (or averaged over a plurality or range of frequencies) is a threshold value (or greater than or equal to the threshold value).

[0032] In various implementations, the particular frequency is a default frequency. In various implementations, the default frequency is based on user input, e.g., setting or changing the default frequency.

[0033] As described above, the primary electronic device **110** transforms the one or more sets of coordinates into the environment coordinate system into locations on the primary display **111** using a transform based on the primary camera pose and displays the virtual pick-up representation **161** at those location on the primary display **111**.

[0034] In various implementations, the primary device **110** provides feedback regarding a target combined audio pick-up pattern. The target combined audio pick-up pattern may be based on the number and/or device types of the primary device and/or secondary devices. In various implementations, the primary device **110** provides haptic feedback at the primary device **110** in proportion (or inverse proportion) to a distance from a location at which the target combined pick-up pattern is achieved. In various implementations, the primary device **110** display, in the primary display **111**, a target representation **162** of the target combined audio pick-up pattern.

[0035] FIG. **1B** illustrates a first physical environment **100** at a second time subsequent to the first time. At the second time, the first secondary electronic device **120A** has moved, moving the first

secondary microphone **121A** from the first secondary microphone pose to an updated first secondary microphone pose. In response to movement of the first secondary electronic device **120A**, the first secondary electronic device representation **151A** has moved in the first XR environment **140** displayed on the primary display **111** of the primary electronic device **110**. Further, in response to the change in microphone pose from the first secondary microphone pose to the updated first secondary microphone pose, the combined audio pick-up pattern and corresponding virtual pick-up pattern representation **161** has changed in the first XR environment **140** displayed on the primary display **111** of the primary electronic device **110**.

[0036] FIG. **1C** illustrates the first physical environment **100** at a third time subsequent to the second time. At the first time, the second time, and additional times, the primary electronic device **110** and the secondary electronic devices **120A-120B** record sound emitted by the cricket **130**.

[0037] At each of the times and each of a plurality of frequencies, a base volume of sound emitted by the cricket **130** can be determined based on the volume of the combined sound at the time and frequency. In various implementations, a true volume of sound emitted by the cricket **130** can be determined based on the base volume of sound emitted by the cricket **130** and the combined audio pick-up pattern at the location of the cricket **130**.

[0038] For each of the microphones, times, and frequencies, the directional volume of sound emitted by the cricket **130** at the time and frequency in the direction of the microphone can be determined based on the volume of sound recorded by the microphone at the time, the pose of the microphone, and the audio pick-up pattern of the microphone.

[0039] In various implementations, the directional volume of sound at each time and frequency is normalized by dividing by the base volume of sound at the time and frequency. In various implementations, values for the normalized directional volume of sound are combined to determine a time-varying, frequency-dependent audio emission pattern for the cricket **130**. In various implementations, the time-varying, frequency-dependent audio emission pattern is averaged over time and/or frequency.

[0040] In FIG. **1C**, the virtual pick-up representation **161** is replaced with a virtual emission representation **163** of the audio emission pattern.

[0041] FIG. **2A** illustrates a second XR environment **200** based on a second physical environment at a first time from a user perspective. The second XR environment **200** includes a table representation **211** of a physical table in the second physical environment and a lamp representation **212** of a physical lamp in the second physical environment. The second XR environment **200** includes virtual flowers **221** as a world-locked virtual object on the table representation **211**. The second XR environment **200** includes a virtual clock **222** as a display-locked virtual object. The second XR environment **200** includes a virtual cricket **223** as a world-locked virtual object on the table representation **221**.

[0042] The virtual cricket **223** is an audio emitter object associated with an audio emission pattern. In various implementations, the audio emission pattern of the virtual cricket **223** is based on the audio emission pattern of the cricket **130** determined in the first physical environment **100**.

Accordingly, in various implementations, the audio emission pattern of the virtual cricket **223** is time-varying and/or frequency-dependent. In various implementations, the sound produced by the virtual cricket **223** is based on the combined sound determined in the first physical environment **100**. In various implementations, the sound produced by the virtual cricket **223** is a different sound, but still is based on the audio emission pattern of the cricket **130**. For example, the audio emission pattern of a physical trumpet playing a first melody in a physical environment can be used to render a virtual trumpet playing a second melody in an XR environment. As another example, the audio emission pattern of a physical person speaking a first set of words can be used to render a virtual person speaking a second set of words.

[0043] The second XR environment **200** includes a volume meter **250** indicating the volume of audio played at the user location at various frequencies. In particular, at the first time, the volume at

a first frequency, **f1**, is a first volume, **V1**, and the volume at a second frequency, **f2**, is a second volume, **V2**. In various implementations, the volume meter **250** is not displayed. However, for ease of explanation, the volume meter **250** is illustrated in FIGS. **2A** and **2B**.

[0044] FIG. **2B** illustrates the second XR environment **200** at a second time from a user perspective. In FIG. **2B**, the virtual cricket **223** has changed pose. Because the virtual cricket **223** has changed pose, the volume of audio played at the user location at the second time is less than the volume of audio played at the user location at the first time. Thus, the sound is quieter or less intense. Further, the change in volume is frequency-dependent such that the change in volume is greater at higher frequencies than at lower frequencies. Accordingly, the second XR environment **200** includes the volume meter **250** indicating the volume of audio played at the second user location at the first frequency, **f1**, is a second volume, **V2**, less than the first volume, **V1**, and the volume of audio played at the second user location at the second frequency, **f2**, is a third volume, **V3**, less than the first volume, **V1**, and also less than the second volume, **V2**. Thus, the difference between the first volume, **V1**, and the second volume, **V2**, is less than the difference between the first volume, **V2**, and the third volume, **V3**.

[0045] FIG. **3** is a flowchart representation of a method **300** of visualizing a combined audio pick-up pattern in accordance with some implementations. In various implementations, the method **300** is performed by a first device in a physical environment, the first device having a display, one or more processors, and non-transitory memory. In some implementations, the method **300** is performed by processing logic, including hardware, firmware, software, or a combination thereof. In some implementations, the method **300** is performed by a processor executing instructions (e.g., code) stored in a non-transitory computer-readable medium (e.g., a memory).

[0046] The method **300** begins, in block **310**, with the device determining a first audio pick-up pattern of the first device. In various implementations, the device determines the first audio pick-up pattern by reading data from non-transitory memory indicative of the first audio pick-up pattern of the first device.

[0047] The method **300** continues, in block **320**, with the device determining one or more second audio pick-up patterns of a respective one or more second devices. In various implementations, the first device determines the second audio pick-up patterns by detecting the second devices in an image of the physical environment, classifying each of the second devices as a device type, and reading data from non-transitory memory (local or remote) associating device types with audio pick-up patterns. Thus, in various implementations, determining a particular audio pick-up pattern of a particular second device includes determining a device type of the particular second device. In various implementations, the first device determines the second audio pick-up patterns by receiving data from the second electronic devices. In various implementations, the first device determines the second audio pick-up patterns by reading machine-readable code displayed by the second electronic devices.

[0048] The method **300** continues, in block **330**, with the device determining a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns.

[0049] In various implementations, determining the combined audio pick-up pattern includes converting at least one of the first audio pick-up pattern and the one or more second audio pick-up patterns into a global Cartesian audio pick-up pattern. Accordingly, in various implementations, determining at least one of the first audio pick-up pattern or the one or more second audio pick-up patterns includes converting a polar audio pick-up pattern into a local Cartesian audio pick-up pattern based on an inverse-square law. Further, in various implementations, determining at least one of the first audio pick-up pattern or the one or more second audio pick-up patterns includes converting a local Cartesian audio pick-up pattern into a global Cartesian audio pick-up pattern based on a location and/or orientation of the associated device. Thus, in various implementations, determining a particular audio pick-up pattern of a particular device includes determining a

location and/or orientation of the particular device in the physical environment. In various implementations, the particular device is the first device. In various implementations, the first device determines the location and/or orientation of the first device using an IMU of the first device.

[0050] In various implementations, the particular device is a particular second device. In various implementations, the first device determines the location and/or orientation of the particular second device by detecting the particular second device in the image of the physical environment. In various implementations, the first device determines the location and/or orientation of the particular second device by receiving pose data from the particular second device. In various implementations, the first device determines the location and/or orientation of the particular second device by reading machine-readable code displayed by the particular second device in the image of the physical environment.

[0051] In various implementations, determining the combined audio pick-up pattern includes combining (e.g., adding) the first audio pick-up pattern and the one or more second audio pick-up patterns.

[0052] The method **300** continues, in block **340**, with the device displaying, on the display, a representation of the combined audio pick-up pattern. In various implementations, the first device determines one or more sets of coordinates in the coordinate system of the physical environment for the representation of the combined audio pick-up pattern at which the combined audio pick-up pattern is a threshold value. Thus, in various implementations, the representation of the combined audio pick-up pattern is a surface including locations in the physical environment at which the combined pick-up level is constant. In various implementations, displaying the representation of the combined audio pick-up pattern includes displaying the representation of the combined audio pick-up pattern in association with an image of the physical environment at a location in the physical environment (e.g., as a world-locked virtual object).

[0053] In various implementations, the method **300** includes detecting movement of at least one of the first device or the one or more second devices. The method **300** includes determining an updated combined audio pick-up pattern of the first device and the one or more second devices based on the movement. The method **300** includes displaying, on the display, a representation of the updated combined audio pick-up pattern.

[0054] In various implementations, the method **300** includes providing feedback regarding a target combined audio pick-up pattern. The target combined audio pick-up pattern may be based on the number and/or device types of the first device and/or one or more second devices. In various implementations, providing feedback regarding the target combined audio pick-up pattern includes providing haptic feedback at the first device in proportion to a distance from a location at which the target combined pick-up pattern is achieved. In various implementations, providing feedback regarding the target combined audio pick-up pattern includes displaying a representation of the target combined audio pick-up pattern.

[0055] As described above, in various implementations, the sound recorded by multiple microphones of multiple devices is combined to generate a combined sound. The combined sound can be used to determine a combined audio pick-up pattern of the multiple devices. Further, the combined sound can be used to determine an audio emission pattern of a sound source.

[0056] FIG. **4** is a flowchart representation of a method **400** of determining an audio emission pattern in accordance with some implementations. In various implementations, the method **400** is performed by a first device at a first location, the first device having a microphone, one or more processors, and non-transitory memory. In some implementations, the method **400** is performed by processing logic, including hardware, firmware, software, or a combination thereof. In some implementations, the method **400** is performed by a processor executing instructions (e.g., code) stored in a non-transitory computer-readable medium (e.g., a memory).

[0057] The method **400** begins, in block **410**, with the first device obtaining, via the microphone,

first audio of a sound source.

[0058] The method **400** continues, in block **420**, with the first device receiving, from one or more second devices, one or more second audio of the sound source.

[0059] The method **400** continues, in block **430**, with the first device determining one or more second locations of the one or more second devices. In various implementations, the first device determines the location and/or orientation of a particular second device by detecting the particular second device in one or more images of a physical environment. In various implementations, the first device determines the location and/or orientation of a particular second device by receiving pose data from the particular second device. In various implementations, the first device determines the location and/or orientation of a particular second device by reading machine-readable code displayed by the particular second device in the image of the physical environment.

[0060] The method **400** continues, in block **440**, with the first device determining an audio emission pattern of the sound source based on the first audio, the one or more second audio, and the one or more second locations, wherein the audio emission pattern of the sound source indicates a sound level at various locations relative to the sound source.

[0061] In various implementations, determining the audio emission pattern includes determining combined audio based on the first audio and the one or more second audio. In various implementations, determining the combined audio includes synchronizing the first audio and the one or more second audio. In various implementations, the first audio and the one or more second audio are synchronized using cross-correlations. In various implementations, determining the combined audio includes filtering direct audio from reverberation audio in at least one of the first audio and one or more second audio. In various implementations, the filtering includes zero-forcing or MMSE (“minimum mean squared-error”) equalization. In various implementations, determining the combined audio includes performing noise reduction in at least one of the first audio and one or more second audio.

[0062] In various implementations, determining the audio emission pattern includes determining a base volume of the sound source at each of a plurality of times and each of a plurality of frequencies. In various implementations, the base value of the sound source at each of the plurality of times and each of the plurality of frequencies is determined based on the volume of the combined audio at the time and frequency.

[0063] In various implementations, determining the audio emission pattern includes, for the first audio (associated with the first device) and each of the one or more second audio (respectively associated with the one or more second devices), determining a directional volume of the sound source at each of the plurality of times and each of the plurality of frequencies. In various implementations, determining the directional volume of the sound source at each of the plurality of times and each of the plurality of frequencies is based on a volume of the audio at the time, a pose of the respective device, and an audio pick-up pattern of the respective device.

[0064] In various implementations, determining the audio emission pattern includes, for the first audio and each of the one or more second audio, determining a normalized directional volume of sound at each of the plurality of times and each of the plurality of frequencies by dividing the directional volume of sound at the time and frequency by the base volume of sound at the time and frequency.

[0065] In various implementations, the normalized directional volume of sound for the first audio and each of the one or more second audio at each of the plurality of times and each of the plurality of frequencies are combined to generate a time-varying, frequency-dependent audio emission pattern. Thus, in various implementations, the audio emission pattern of the sound source indicates, at a particular location relative to the sound source, a sound level at various frequencies. Further, in various implementations, the audio emission pattern of the sound source indicates, at a particular location relative to the sound source, a sound level at various times. In various implementations, the time-varying, frequency-dependent audio emission pattern is averaged over time and/or

frequency.

[0066] In various implementations, determining the audio emission pattern includes interpolating the normalized directional volume of sound at two locations to determine the normalized directional volume of sound at a different location. Thus, in various implementations, determining the audio emission pattern of the sound source includes determining a sound level at a third location, different from the first location and the one or more second locations, based on at least two of a sound level at the first location and one or more sound levels at the one or more second locations.

[0067] In various implementations, the method **400** includes storing the audio emission pattern of the sound source in association with the combined audio. Thus, the audio emission pattern of the sound source can be used to render a virtual sound source playing the combined audio. In various implementations, the audio emission pattern of the sound source can be used to render a virtual sound source playing audio other than the combined audio.

[0068] FIG. **5** is a block diagram of an electronic device **500** in accordance with some implementations. While certain specific features are illustrated, those skilled in the art will appreciate from the present disclosure that various other features have not been illustrated for the sake of brevity, and so as not to obscure more pertinent aspects of the implementations disclosed herein. To that end, as a non-limiting example, in some implementations the electronic device **500** includes one or more processing units **502** (e.g., microprocessors, ASICs, FPGAs, GPUs, CPUs, processing cores, and/or the like), one or more input/output (I/O) devices and sensors **506**, one or more communication interfaces **508** (e.g., USB, FIREWIRE, THUNDERBOLT, IEEE 802.3x, IEEE 802.11x, IEEE 802.16x, GSM, CDMA, TDMA, GPS, IR, BLUETOOTH, ZIGBEE, and/or the like type interface), one or more programming (e.g., I/O) interfaces **510**, one or more XR displays **512**, one or more optional interior- and/or exterior-facing image sensors **514**, a memory **520**, and one or more communication buses **504** for interconnecting these and various other components.

[0069] In some implementations, the one or more communication buses **504** include circuitry that interconnects and controls communications between system components. In some implementations, the one or more I/O devices and sensors **506** include at least one of an inertial measurement unit (IMU), an accelerometer, a gyroscope, a thermometer, one or more physiological sensors (e.g., blood pressure monitor, heart rate monitor, blood oxygen sensor, blood glucose sensor, etc.), one or more microphones, one or more speakers, a haptics engine, one or more depth sensors (e.g., a structured light, a time-of-flight, or the like), and/or the like.

[0070] In some implementations, the one or more XR displays **512** are configured to present XR content to the user. In some implementations, the one or more XR displays **512** correspond to holographic, digital light processing (DLP), liquid-crystal display (LCD), liquid-crystal on silicon (LCoS), organic light-emitting field-effect transitory (OLET), organic light-emitting diode (OLED), surface-conduction electron-emitter display (SED), field-emission display (FED), quantum-dot light-emitting diode (QD-LED), micro-electro-mechanical system (MEMS), and/or the like display types. In some implementations, the one or more XR displays **512** correspond to diffractive, reflective, polarized, holographic, etc. waveguide displays. For example, the electronic device **500** includes a single XR display. In another example, the electronic device **500** includes an XR display for each eye of the user. In some implementations, the one or more XR displays **412** are capable of presenting AR, MR, and/or VR content.

[0071] In various implementations, the one or more XR displays **512** are video passthrough displays which display at least a portion of a real scene as an image captured by a scene camera. In various implementations, the one or more XR displays **512** are optical see-through displays which are at least partially transparent and pass light emitted by or reflected off the real scene.

[0072] In some implementations, the one or more image sensors **514** are configured to obtain image data that corresponds to at least a portion of the face of the user that includes the eyes of the

user (any may be referred to as an eye-tracking camera). In some implementations, the one or more image sensors **514** are configured to be forward-facing so as to obtain image data that corresponds to the physical environment as would be viewed by the user if the electronic device **500** was not present (and may be referred to as a scene camera). The one or more optional image sensors **514** can include one or more RGB cameras (e.g., with a complimentary metal-oxide-semiconductor (CMOS) image sensor or a charge-coupled device (CCD) image sensor), one or more infrared (IR) cameras, one or more event-based cameras, and/or the like.

[0073] The memory **520** includes high-speed random-access memory, such as DRAM, SRAM, DDR RAM, or other random-access solid-state memory devices. In some implementations, the memory **520** includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid-state storage devices. The memory **520** optionally includes one or more storage devices remotely located from the one or more processing units **502**. The memory **520** comprises a non-transitory computer readable storage medium. In some implementations, the memory **520** or the non-transitory computer readable storage medium of the memory **520** stores the following programs, modules and data structures, or a subset thereof including an optional operating system **530** and an XR presentation module **540**.

[0074] The operating system **530** includes procedures for handling various basic system services and for performing hardware dependent tasks. In some implementations, the XR presentation module **540** is configured to present XR content to the user via the one or more XR displays **512**. To that end, in various implementations, the XR presentation module **540** includes a data obtaining unit **542**, an audio pattern determining unit **544**, an XR presenting unit **546**, and a data transmitting unit **548**.

[0075] In some implementations, the data obtaining unit **542** is configured to obtain data (e.g., presentation data, interaction data, sensor data, location data, etc.). The data may be obtained from the one or more processing units **502** or another electronic device. To that end, in various implementations, the data obtaining unit **542** includes instructions and/or logic therefor, and heuristics and metadata therefor.

[0076] In some implementations, the audio pattern determining unit **544** is configured to determine a combined audio pick-up pattern and/or an audio emission pattern of a sound source. To that end, in various implementations, the audio pattern determining unit **544** includes instructions and/or logic therefor, and heuristics and metadata therefor.

[0077] In some implementations, the XR presenting unit **546** is configured to present XR content via the one or more XR displays **512**. To that end, in various implementations, the XR presenting unit **546** includes instructions and/or logic therefor, and heuristics and metadata therefor.

[0078] In some implementations, the data transmitting unit **548** is configured to transmit data (e.g., presentation data, location data, etc.) to the one or more processing units **502**, the memory **520**, or another electronic device. To that end, in various implementations, the data transmitting unit **548** includes instructions and/or logic therefor, and heuristics and metadata therefor.

[0079] Although the data obtaining unit **542**, the audio pattern determining unit **544**, the XR presenting unit **546**, and the data transmitting unit **548** are shown as residing on a single electronic device **500**, it should be understood that in other implementations, any combination of the data obtaining unit **542**, the audio pattern determining unit **544**, the XR presenting unit **546**, and the data transmitting unit **548** may be located in separate computing devices.

[0080] Moreover, FIG. 5 is intended more as a functional description of the various features that could be present in a particular implementation as opposed to a structural schematic of the implementations described herein. As recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated. For example, some functional modules shown separately in FIG. 5 could be implemented in a single module and the various functions of single functional blocks could be implemented by one or more functional blocks in

various implementations. The actual number of modules and the division of particular functions and how features are allocated among them will vary from one implementation to another and, in some implementations, depends in part on the particular combination of hardware, software, and/or firmware chosen for a particular implementation.

[0081] While various aspects of implementations within the scope of the appended claims are described above, it should be apparent that the various features of implementations described above may be embodied in a wide variety of forms and that any specific structure and/or function described above is merely illustrative. Based on the present disclosure one skilled in the art should appreciate that an aspect described herein may be implemented independently of any other aspects and that two or more of these aspects may be combined in various ways. For example, an apparatus may be implemented and/or a method may be practiced using any number of the aspects set forth herein. In addition, such an apparatus may be implemented and/or such a method may be practiced using other structure and/or functionality in addition to or other than one or more of the aspects set forth herein.

[0082] It will also be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first node could be termed a second node, and, similarly, a second node could be termed a first node, which changing the meaning of the description, so long as all occurrences of the “first node” are renamed consistently and all occurrences of the “second node” are renamed consistently. The first node and the second node are both nodes, but they are not the same node.

[0083] The terminology used herein is for the purpose of describing particular implementations only and is not intended to be limiting of the claims. As used in the description of the implementations and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0084] As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting,” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” may be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

Claims

1. A method comprising: at a first device in a physical environment, the first device having a display, one or more processors, and non-transitory memory; determining a first audio pick-up pattern of the first device; determining one or more second audio pick-up patterns of a respective one or more second devices; determining a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns; and displaying, on the display, a representation of the combined audio pick-up pattern.

2. The method of claim 1, wherein determining a particular audio pick-up pattern of a particular second device includes determining a device type of the particular second device.

3. The method of claim 1, wherein determining a particular audio pick-up pattern of a particular device includes determining a location and/or orientation of the particular device in the physical environment.
4. The method of claim 3, wherein determining the location and/or orientation of the particular device includes detecting a particular second device in an image of the physical environment.
5. The method of claim 4, wherein determining the location and/or orientation of the particular second device includes reading machine-readable code displayed by the particular second device in the image of the physical environment.
6. The method of claim 1, wherein determining the combined audio pick-up pattern includes converting at least one of the first audio pick-up pattern and the one or more second audio pick-up patterns into a global Cartesian audio pick-up pattern.
7. The method of claim 1, wherein determining the combined audio pick-up pattern includes adding the first audio pick-up pattern and the one or more second audio pick-up patterns.
8. The method of claim 1, wherein displaying the representation of the combined audio pick-up pattern includes displaying the representation of the combined audio pick-up pattern in association with an image of the physical environment at a location in the physical environment.
9. The method of claim 1, wherein the representation of the combined audio pick-up pattern is a surface including locations in the physical environment at which the combined pick-up level is constant.
10. The method of claim 1, further comprising: detecting movement of at least one of the first device or the one or more second devices; determining an updated combined audio pick-up pattern of the first device and the one or more second devices based on the movement; and displaying, on the display, a representation of the updated combined audio pick-up pattern.
11. The method of claim 1, further comprising providing feedback regarding a target combined audio pick-up pattern.
12. The method of claim 11, wherein providing feedback regarding the target combined audio pick-up pattern includes displaying a representation of the target combined audio pick-up pattern.
13. A first device comprising: a display; non-transitory memory; and one or more processors to: determine a first audio pick-up pattern of the first device; determine one or more second audio pick-up patterns of a respective one or more second devices; determine a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns; and display, on the display, a representation of the combined audio pick-up pattern.
14. The first device of claim 13, wherein the one or more processors are to determine a particular audio pick-up pattern of a particular second device by determining a device type of the particular second device.
15. The first device of claim 13, wherein the one or more processors are to determine a particular audio pick-up pattern of a particular device by determining a location and/or orientation of the particular device in the physical environment.
16. The first device of claim 13, wherein the one or more processors are to display the representation of the combined audio pick-up pattern by displaying the representation of the combined audio pick-up pattern in association with an image of the physical environment at a location in the physical environment.
17. The first device of claim 13, wherein the one or more processors are further to: detect movement of at least one of the first device or the one or more second devices; determine an updated combined audio pick-up pattern of the first device and the one or more second devices based on the movement; and display, on the display, a representation of the updated combined audio pick-up pattern.
18. The first device of claim 13, wherein the one or more processors are further to provide feedback regarding a target combined audio pick-up pattern.

19. The first device of claim 18, wherein the one or more processors are to provide feedback regarding the target combined audio pick-up pattern by displaying a representation of the target combined audio pick-up pattern.

20. A non-transitory memory storing one or more programs, which, when executed by one or more processors of a device including a microphone, cause the device to: determine a first audio pick-up pattern of the first device; determine one or more second audio pick-up patterns of a respective one or more second devices; determine a combined audio pick-up pattern of the first device and the one or more second devices based on the first audio pick-up pattern and the one or more second audio pick-up patterns; and display, on the display, a representation of the combined audio pick-up pattern.
